The main reference of this study log will be around the paper Towards Observability Data Management at Scale which focuses on Slack's crisis management when their core systems experience critical failure. For me, this shows that observability is always one of the foremost requirements to run a large scale system. While the cost to maintain this consistent logging and monitoring can affect performance, the downside of not having this system in place is catastrophic. One thing I would like to note though is that for purely cloud-based systems, this monitoring can be organized relatively safely behind firewalls. But if we look at edge/IoT computing where small devices are connected to the internet by default with their own metrics/logs collections methods, is this a safe design? For people working in Operation Technology (OT), this becomes a serious security issue. Some of these components are interacting with older devices daily, devices that do not have any sort of monitoring and security features. This means attackers can move laterally freely and hide themselves in some of these OT devices. Before industry's standards move to things like software-defined networking or microsegmentation, these can always be potential security threats.

As for which components/steps of the observation that are the most important, based off of the paper and my own experience, I would argue the trace/analysis are the key component/step in the observation process. Logs collection usually involves collecting symptoms/configurations from the entire systems and it's up to the support teams to filter through potentially millions of logs lines to reach to the crux of the issue. For me, that means collecting is not as important as analysis. While one can say that if you can't capture the logs and the relevant trace, you wouldn't be able to come up with a good analysis but preliminary analysis can review hidden information that capturing might miss. One of the most hyped use cases for AI agents now involve assisting with the analysis of these logs through the use of things like RAG agents. Visualization is also helpful to form a timeline of events of what has happened but ultimately any good systems at scale should have a robust logging system that can showcase the core issues in the first place.